

Multiple Choice

1. **Answer:** A

Options: A) Decreases model bias; B) Decreases estimation bias; C) Decreases variance; D) Doesn't affect bias and variance

Note: Correct option: A - Decreases model bias

2. **Answer:** B

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: B - False, False

3. **Answer:** C

Options: A) Regularization is too low and model is overfitting; B) Regularization is too high and model is underfitting; C) Step size is too large; D) Step size is too small

Note: Correct option: C - Step size is too large

4. **Answer:** B

Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$

Note: Correct option: B - $O(N)$

5. **Answer:** C

Options: A) 0; B) 1; C) 2; D) 3

Note: Correct option: C - 2

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Improve the optimisation algorithm being used for error minimisation.'

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, False'.

Long Form Response

11. **Answer:** Your decision trees are too shallow.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** `5 + torch.randn(10,5) * 4; 2 * torch.rand(10,10) - 1`. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key